

Thyroglobulin Antibody and Thyroglobulin, IMA or LC/MS- MS

Test Details

METHODOLOGY

TgAb: Beckman Coulter immunometric assay; **Tg-IMA:** Beckman Coulter immunometric assay; **Tg:** Liquid chromatography/tandem mass spectrometry (LC/MS-MS) lab-developed test. Both Tg assays are calibrated against CRM-57 international standard for Tg as recommended in recent guidance documents.¹

LabCorp uses a dual Tg strategy in an effort to eliminate the potential effect of TgAb interference on Tg analysis. All specimens are initially assayed for TgAb by the highly sensitive Beckman Coulter IMA. Samples with TgAb below the detectable limit are assayed for Tg by a sensitive second-generation IMA (Beckman Coulter) with a relatively rapid turnaround time. Specimens with any measurable levels of TgAb are assayed by a sensitive LC/MS-MS method.

RESULT TURNAROUND TIME

1 - 5 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Thyroglobulin antibody (TgAb) testing is performed on all samples. If TgAb is negative, serum thyroglobulin is measured by a sensitive IMA with a relatively rapid turnaround time, generally two to three days. If TgAb is positive, serum thyroglobulin is measured by a sensitive LC/MS-MS method. This method is less prone to interference by TgAb but has a longer turnaround time of four to six days.

Use

Thyroglobulin (Tg) measurement is intended to aid in monitoring for the presence of orthotopic and/or metastatic thyroid tissues in patients who have had thyroid gland ablation (using thyroid surgery with or without radioactivity). Measurement of thyroglobulin antibody (TgAb) is used to identify patient samples that may be affected by TgAb interference in the measurement of Tg. Quantitative TgAb concentrations can also serve as a surrogate tumor marker for DTC recurrence and for monitoring changes in tumor mass in certain patients.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

Any changes in serum Tg concentrations should be interpreted in light of the total clinical presentation of the patient, including clinical history, data from additional testing and other appropriate information. Single measurement of thyroglobulin close to the limit of detection is of minimal value in assessing disease status. Serial determinations are required, and should be referenced to the postsurgical baseline Tg result when possible. Evaluation of increasing Tg levels over time are more clinically important.^{1,2}

Serial thyroglobulin (Tg) and/or thyroglobulin antibody (TgAb) testing on an individual patient should be performed by the same method for reliable interpretation.^{1,3-5} When a change in Tg method is necessary, it is recommended to reestablish a new baseline Tg level to then interpret further change over time.

It should be noted that the thyroid tumors of some patients fail to secrete a detectable Tg concentration or may secrete abnormal Tg isoforms that are not detectable by some assays used to measure Tg.¹ Measurement of preoperative Tg levels can provide the clinician with insights regarding the tumors Tg production and secretion and support the utility of postoperative Tg monitoring.

As with all two-site "sandwich" immunoassays, some of the analytical limitations of Tg IMA include hook effects, human mouse antibody (HAMA), and anti-Tg interference.⁶ The Access Thyroglobulin assay does not demonstrate any "hook" effect for concentrations up to 40,000 ng/mL. The Access Thyroglobulin Antibody does not demonstrate any "hook" effect up to approximately 50,000 IU/mL. For samples that are Tg antibody positive, Tg will be measured using a sensitive LC/MS-MS method that is not subject to TgAb interference.

Rare amino acid sequence mutations within Tg could potentially cause a false-low result in the Tg LC/MS-MS assay, if the sequence variation occurs within the tryptic peptide measured by the assay or eliminates the tryptic cleavage site.⁷ In the heterozygote state, the result would be an apparent reduction in Tg concentration by about 50%, while the homozygous state no TG would be detected.

Custom Additional Information

Thyroglobulin (Tg) is synthesized exclusively by thyroid follicular cells as the precursor protein for thyroid hormone biosynthesis.^{1,5,8} Serum Tg reflects the integrated sum of Tg released into the blood stream and is related to three factors:

1. Release of Tg into the blood stream is proportion to the mass of thyroid tissue present (from both normal and any tumor present).
2. The effect of any thyroid injury caused by inflammation associated with thyroiditis; radioactive iodine (RAI) therapy; fine needle aspiration (FNA) biopsy; or surgery.
3. The degree of TSH-receptor stimulation from endogenous TSH, recombinant human TSH (rhTSH), human chorionic gonadotropin (during pregnancy) or antibodies that stimulate the TSH receptor (Graves' disease).

Measurement of serum Tg is used primarily in the postoperative management and long-term surveillance of patients with DTC. Serum Tg concentrations reflect the residual mass of thyroid tissue present (normal remnant and/or tumor) and the effect of the three factors listed above.^{1,5,7} Since Tg production is thyroid-specific but not tumor-specific, patient-related factors influence the interpretation of serum Tg concentrations. Postoperative serum Tg levels are related to the amount of residual normal and tumor tissue, recent thyroid injury, and the TSH status of the patient. In patients who have not had a total thyroidectomy and radioiodine remnant ablation, the interpretation of serum Tg measurements is limited by the inability to differentiate between tumor and thyroid remnant, though trends over time can be informative. An increase in serum Tg levels over time while on suppressive thyroxine therapy is highly suggestive of tumor recurrence or progression.⁷

Thyroglobulin antibody (TgAb) is detected in approximately 10% of the general population and in 20% of patients with DTC.^{5,9,10} Several studies have suggested that the quantitative TgAb concentrations can serve as a surrogate tumor marker for DTC recurrence and for monitoring changes in tumor mass in certain patients.¹¹

Guidelines recommend that TgAb be measured in all specimens that are tested for Tg because TgAb interference with Tg-IMA measurements is associated with underestimation of Tg and the potential for reporting inappropriately low or undetectable values that can disguise the presence of disease.^{1,7,8,12-15}

Historically, there have been no reliable methods for the accurate measurement of Tg in patients with TgAb; however, new assays employing trypsin digestion of serum proteins followed by LC-MS/MS quantitation of Tg-specific tryptic peptides allow for the accurate quantitation of Tg in the presence of TgAb.⁶

Reference ranges established by testing normal, euthyroid subjects have little relevance when interpreting serum Tg concentrations in differentiated thyroid cancer (DTC) patients after thyroidectomy. Current guidelines recommend using a given assay's functional sensitivity (or LOQ) as the clinical threshold for distinguishing biochemically negative patients from those with residual Tg-producing tissue.⁸

- The Beckman Coulter IMA Tg assay (used for TgAb-negative samples) has a functional sensitivity/limit of quantitation (LOQ) of 0.1 ng/mL.
- The LC/MS-MS Tg assay (used for TgAb-positive patients) has a functional sensitivity/limit of quantitation (LOQ) of 0.2 ng/mL.

For the detection of interfering TgAb, the lower limit of detection (LOD) of the TgAb assay should be used. TgAb interference with Tg-IMA assays can occur at low levels of TgAb.

- The Beckman Coulter TgAb assay has a limit of detection (LOD) of 0.9 IU/mL.

Specimen Requirements

Specimen

Serum

Volume

3 mL (two tubes, 1.5 mL each tube)

Minimum Volume

2 mL (two tubes, 1 mL each tube) (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube or gel-barrier tube

Collection Instructions

Separate serum from cells, and transfer specimen to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	7 days
Refrigerated	7 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Storage Instructions

Room temperature

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Footnotes

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